AI for Medicine – Project Report

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1. Project Title

Heart Disease Prediction from Clinical Parameters Using Machine Learning Models and Nested Cross-Validation

2. Problem Statement

Cardiovascular diseases (CVDs) are among the leading causes of mortality worldwide, making early diagnosis and risk assessment essential components of modern healthcare. Identifying patients at high risk can improve clinical outcomes, support preventive interventions, and reduce healthcare costs.

Machine learning has emerged as a valuable tool for analyzing clinical data and supporting medical decision-making. By learning patterns from previously observed patient records, machine learning models can assist clinicians in identifying individuals who may be affected by a disease.

This project investigates whether routinely collected clinical variables can be used to predict the presence of heart disease. Two machine learning approaches, Logistic Regression and Random Forest, were developed and compared using rigorous validation procedures designed to prevent data leakage and provide reliable estimates of model performance.

3. Objective of the Study

The objective of this study is to develop and evaluate machine learning models capable of predicting the presence or absence of heart disease using patient clinical data.

More specifically, the project aims to:

1. Build classification models using clinical variables.
2. Compare Logistic Regression and Random Forest classifiers.
3. Evaluate model performance using robust validation techniques.
4. Identify the model that provides the best discrimination between healthy and diseased patients.
5. Assess whether threshold adjustment can improve sensitivity for screening purposes.

4. Dataset Description

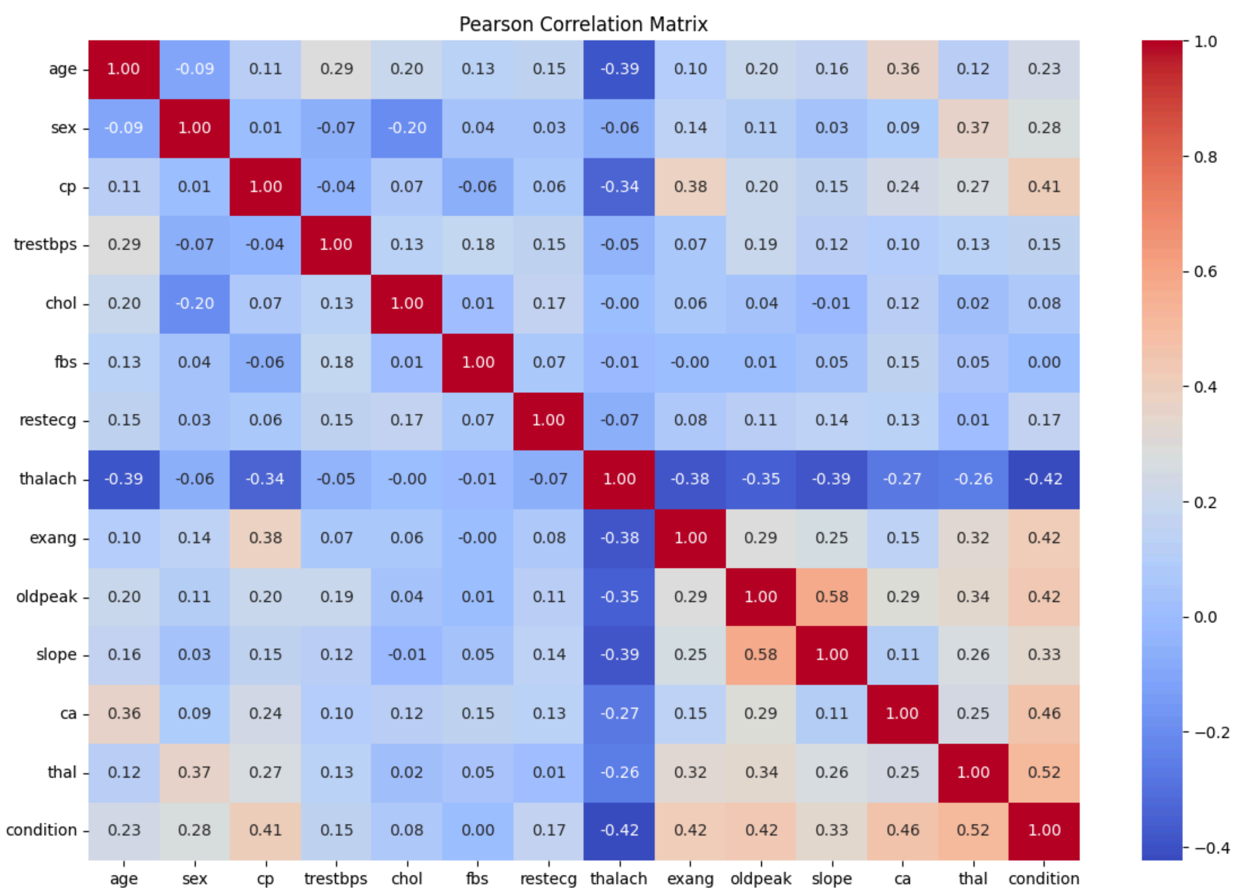
The dataset consists of **297 tabular samples**, where each row represents a patient and columns correspond to clinical variables, including demographic, physiological, and diagnostic features. The dataset contains **13 input features** and one target variable (*condition*), which indicates the presence of heart disease:

- 0 → No disease
- 1 → Presence of disease

The class distribution includes **160 healthy patients** and **137 patients with heart disease**, resulting in a relatively balanced dataset. Therefore, no severe class imbalance issues are observed. An exploratory data analysis confirmed that there are no missing values or duplicated records in the dataset, and no significant noise issues were detected.

5. Data Preprocessing

The dataset was inspected for missing values, duplicates, and inconsistencies, and exploratory data analysis was conducted to examine feature distributions and relationships. No issues were identified.



The correlation matrix was used to investigate relationships among clinical variables and to identify potential multicollinearity issues. No severe correlations requiring feature removal were observed.
The dataset was divided into input features (X) and target labels (y). An 80/20 stratified train-test split was performed to preserve class proportions.
For Logistic Regression, a preprocessing pipeline was implemented using a ColumnTransformer. Numerical variables were standardized using StandardScaler, while categorical variables were transformed using OneHotEncoder. These operations were incorporated directly within a Scikit-Learn pipeline.
Random Forest did not require feature scaling because tree-based algorithms are generally insensitive to feature magnitude.
Special attention was paid to preventing data leakage. All preprocessing operations were learned exclusively from training data during cross-validation, ensuring that information from validation and test sets could not influence model training.
Nested cross-validation was employed using a 5-fold outer loop and a 3-fold inner loop. The inner loop was used for hyperparameter optimization, whereas the outer loop provided an unbiased estimate of model performance.

6. Avoiding Data Leakage

Preventing data leakage was a key consideration throughout this study. The dataset was first divided into training and testing subsets using an 80/20 stratified split, and the test set was used only after model selection.
For Logistic Regression, all preprocessing operations, including feature scaling and categorical encoding, were incorporated within Scikit-Learn pipelines. This ensured that preprocessing parameters were learned exclusively from the training data during each cross-validation fold.
Nested cross-validation (5-fold outer loop, 3-fold inner loop) was used to separate hyperparameter tuning from model evaluation and obtain unbiased performance estimates.
Additionally, each observation corresponds to a single patient, with no repeated acquisitions or patient identifiers present in the dataset, reducing the risk of patient-level leakage.

7. Machine Learning Pipeline

Two machine learning algorithms were evaluated in this study: Logistic Regression (LR) and Random Forest (RF). These models were selected because they represent two fundamentally different approaches to classification. Logistic Regression is a linear probabilistic model that provides interpretable coefficients, whereas Random Forest is an ensemble learning method capable of capturing complex nonlinear relationships.

Logistic Regression

Logistic Regression estimates the probability that a patient belongs to the positive class (heart disease) using the logistic function.
The model was implemented within a Scikit-Learn pipeline that included:

1. Numerical feature standardization using StandardScaler.
2. Categorical feature encoding using OneHotEncoder.

3. Logistic Regression classification.

Hyperparameter optimization was performed using GridSearchCV within the inner cross-validation loop.

The following hyperparameters were explored:

- Regularization strength (C)
- Class weighting strategies
- Solver configurations

The inclusion of class weighting was investigated to evaluate whether performance could be improved by assigning greater importance to the minority class.

Random Forest

Random Forest is an ensemble algorithm that combines predictions from multiple decision trees. By aggregating many trees trained on bootstrapped samples of the dataset, Random Forest generally achieves greater robustness and lower variance than individual decision trees.

The Random Forest classifier was optimized through hyperparameter tuning of:

- Number of trees (n_estimators)
- Maximum tree depth
- Minimum samples required for node splitting
- Class weighting options

Because tree-based methods are not sensitive to feature scaling, no explicit preprocessing stage was required.

Model Evaluation Framework

Performance evaluation was conducted using nested stratified cross-validation.

The evaluation procedure consisted of:

1. Hyperparameter optimization in the inner loop.
2. Independent model assessment in the outer loop.
3. Selection of the best-performing model based on ROC-AUC.
4. Final evaluation on a completely unseen test set.

The following performance metrics were calculated:

- Accuracy
- Precision
- Recall (Sensitivity)
- F1-score
- Specificity
- ROC-AUC

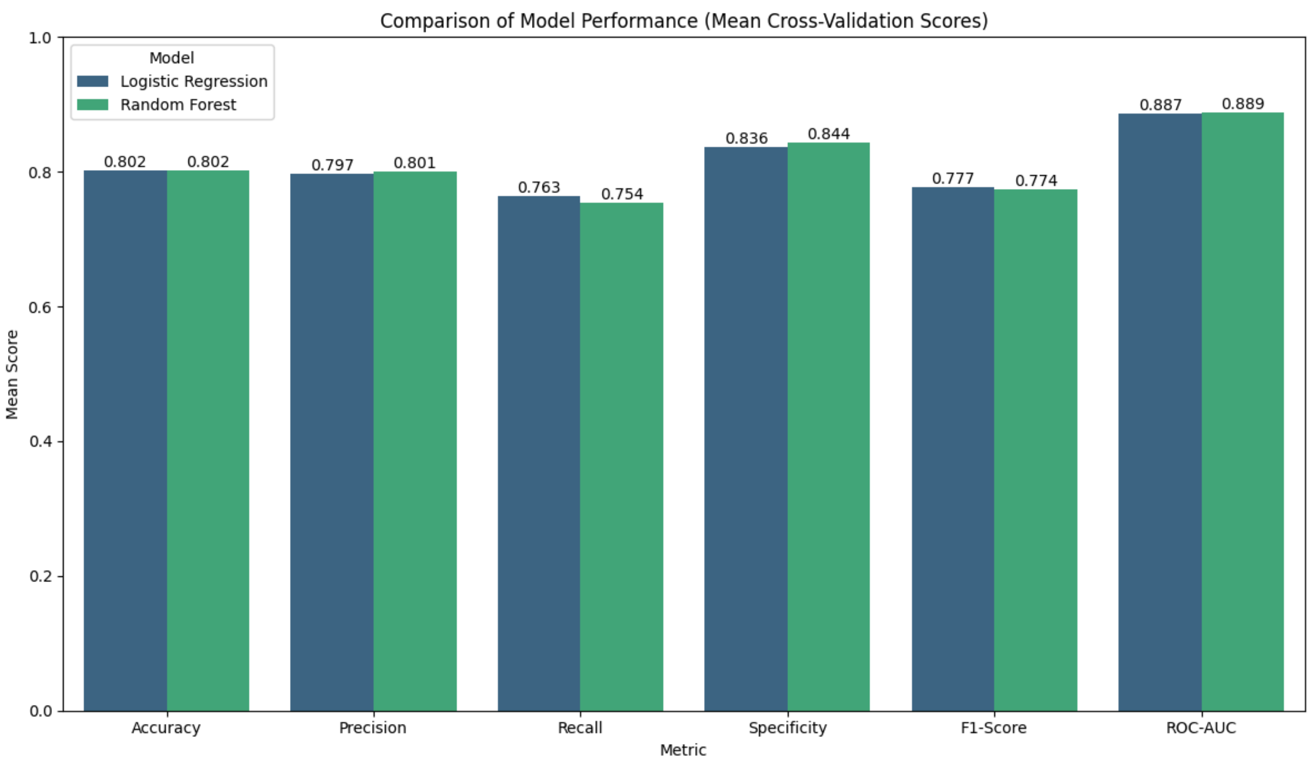
Among these metrics, ROC-AUC was selected as the primary criterion for model comparison because it is independent of a specific decision threshold and provides a comprehensive measure of discriminative ability.

8. Results

Nested Cross-Validation Performance

The performance of Logistic Regression and Random Forest was evaluated using nested cross-validation. The mean scores across the outer folds are summarized in this table.

Model	Accuracy	Precision	Recall	Specifity	F1-score	ROC-AUC
Logistic Regression	0.802	0.797	0.763	0.836	0.777	0.887
Random Forest	0.802	0.801	0.754	0.844	0.774	0.889



Both models achieved very similar performance across all evaluation metrics. Random Forest obtained slightly higher ROC-AUC, Precision and specifity while Logistic Regression demonstrated marginally higher Recall and F1-score.

Test Set Performance

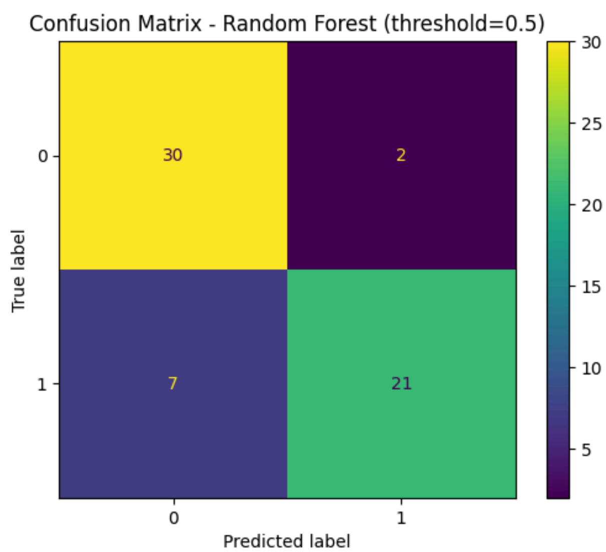
The selected Random Forest model was evaluated on an independent test set. The results are reported in this table.

Metric	Value
Accuracy	0.850
Precision	0.913
Recall	0.750
Specificity	0.938
F1-score	0.824
ROC-AUC	0.940

The model demonstrates strong generalization performance, with particularly high precision and excellent discriminative ability (ROC-AUC ≈ 0.94).
The model also achieved high specificity (93.8%), indicating a strong ability to correctly identify healthy patients.

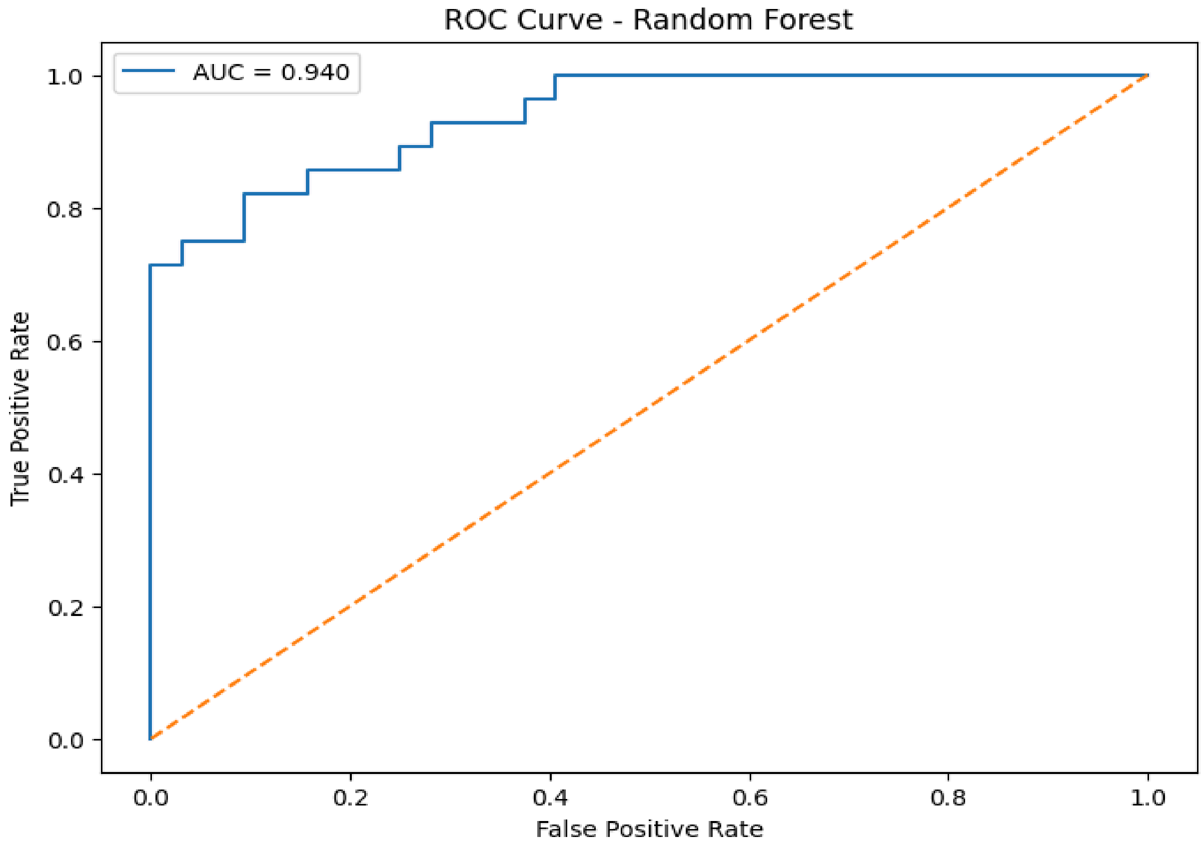
Confusion Matrix

The confusion matrix provides a detailed view of classification performance, including correctly and incorrectly classified samples.



ROC Curve

The ROC curve illustrates the trade-off between the true positive rate and false positive rate across different thresholds.



Threshold Analysis

To further analyze the trade-off between sensitivity and specificity, performance was evaluated at different classification thresholds.

Threshold	Precision	Recall	Specificity	F1-score
0.3	0.683	1.000	0.594	0.812
0.4	0.800	0.857	0.812	0.828
0.5	0.913	0.750	0.938	0.824
0.6	1.000	0.679	1.000	0.809

Lower thresholds increase sensitivity at the expense of specificity, while higher thresholds lead to the opposite effect. This trade-off is particularly relevant in medical decision-making.

9. Discussion

Both Logistic Regression and Random Forest achieved strong and highly comparable predictive performance across all evaluation metrics. While Random Forest obtained a slightly higher ROC-AUC (0.889 vs. 0.887), Logistic Regression showed marginally better recall (0.763 vs. 0.754) and F1-score (0.777 vs. 0.774), indicating similar overall predictive capability. The selection of Random Forest as the final model is consistent with the experimental framework, which uses ROC-AUC as the primary metric for model selection. In addition, Random Forest can capture non-linear relationships and provides feature importance measures, enhancing interpretability. The model also achieved high specificity (93.8%), demonstrating strong ability to correctly identify healthy patients. From a clinical perspective, the slightly higher recall of Logistic Regression may be relevant in screening scenarios where minimizing missed cases is critical, while threshold analysis further highlights the importance of selecting an appropriate operating point depending on the clinical objective.

10. Conclusions and Future Work

This study investigated the use of machine learning techniques for heart disease prediction based on routinely collected clinical variables. Both Logistic Regression and Random Forest demonstrated strong predictive performance, indicating that meaningful clinical patterns can be extracted from the available data.

The comparison between the two models showed highly similar results across all evaluation metrics. Random Forest achieved the highest ROC-AUC and was therefore selected as the final model, while Logistic Regression provided slightly higher recall and comparable F1-score. These findings suggest that both linear and non-linear approaches are suitable for this task, with the final choice depending on the predefined evaluation criterion.

Overall, the results highlight the effectiveness of machine learning for supporting heart disease prediction, while also emphasizing the importance of carefully selecting evaluation metrics and decision thresholds depending on the intended application.

Future work could focus on improving model performance and generalizability by exploring more advanced machine learning algorithms, such as gradient boosting methods (e.g., XGBoost or LightGBM). In addition, alternative model selection strategies based on clinically relevant metrics, such as recall, could be investigated, particularly for screening scenarios where minimizing missed positive cases is critical.

Another important direction is the validation of the proposed models on larger and more diverse datasets, including external cohorts from independent hospitals, in order to assess generalizability. Finally, the integration of explainable artificial intelligence techniques, such as SHAP or LIME, could provide more detailed insights into model predictions and enhance clinical interpretability.

11. Ethics and Data Privacy

The dataset used in this project is publicly available through academic repositories for educational and research purposes.

No personally identifiable information was included in the dataset. The data were already anonymized before publication.

This project did not involve direct patient interaction, clinical intervention, or access to protected health information.

All analyses were conducted solely for academic purposes.

Any third-party software libraries used (Scikit-Learn, NumPy, Pandas, Matplotlib, and Seaborn) are open-source and distributed under their respective licenses.

Researchers using this dataset should nevertheless remain aware that medical datasets may contain demographic biases that can affect fairness and generalizability.

12. Code and Reproducibility

- Core Execution: The project was implemented in a Python Jupyter Notebook (Heart_Disease_ML.ipynb) and executed using Google Colab.
- Software Environment: The experiments were conducted using Python 3.12.13 and standard scientific libraries, including NumPy (2.0.2), Pandas (2.2.2), Scikit-Learn (1.6.1), Matplotlib, and Seaborn.
- Reproducibility Measures: To ensure consistent and reproducible results, a fixed random seed (seed = 42) was applied across all experiments. A stratified train-test split and nested cross-validation framework (5 outer folds, 3 inner folds) were used to provide robust and unbiased performance estimates.
- Pipeline Integrity: All preprocessing steps (feature scaling and encoding) and model training procedures were implemented within Scikit-Learn pipelines. This approach ensures that transformations are learned exclusively from training data, preventing data leakage.
- Data Access and Execution: The experiments can be reproduced by running the notebook in Google Colab, provided that the dataset file (heart_cleveland_upload.csv) is available in the working directory.

13. References

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